Project Title
SmartBuyHub — A Smart Recommendation and Wishlist Web App

Concept Description
SmartBuyHub is a web application that allows users to explore and discover products from an external source, save their favorite items, and receive smart recommendations. The main idea is to provide a seamless and intelligent shopping experience using data from a public product API. Users can log in to create personalized wishlists and view recommended products based on their preferences. The application is targeted toward general online shoppers who are interested in discovering new products with the help of intelligent suggestions.

The core features include:

- Displaying products fetched from a third-party API
- Allowing users to "favorite" items
- Recommending similar products based on past activity
- Saving user data and favorites in a custom backend database

This app focuses on usability and relevance by offering a familiar, yet enhanced shopping interface with a personal touch.

API Integration Plan
We will use the **Fake Store API** (https://fakestoreapi.com), which provides fake product data ideal for prototyping shopping applications.

- **Purpose of the API**: Retrieve product data (e.g., electronics, clothing, jewelry) to display on the web app.
- **Data retrieved**: Titles, prices, descriptions, categories, and product images.
- **Integration method**:
    - The frontend will request product data from our backend server.
    - The backend server will fetch and preprocess API data, then send relevant parts to the frontend.
    - Favorites and browsing logs will be stored in our custom database.

Custom Database Plan
We will build a database that stores user information and product interactions.

Key Tables:

- **Users**: user_id, username, email, password_hash
- **Favorites**: favorite_id, user_id, product_id, timestamp
- **Browse_History**: browse_id, user_id, product_id, timestamp
- **Recommendations**: recommendation_id, user_id, product_id, reason

Data Relationships:

- One user can have many favorites and browsing records.
- Recommendations are generated based on user activity.

This database complements the API data by adding personalization and persistence.

User Service Description
Users can register and log in to the system. Once logged in, they can browse the product catalog (from the API), favorite items, and receive recommended products based on their browsing and favoriting habits. The platform enhances the shopping experience by making it more personal and interactive.

Team Collaboration Plan

- **Team Members**:
  - **Member A** – Backend Developer: API integration, server logic, database design.
  - **Member B** – Frontend Developer: UI/UX design, displaying API data, user interaction.
- **Communication**:
  - Weekly meetings on Microsoft Teams
  - Ad-hoc communication on Slack
- **Collaboration Tools**:
  - **GitHub** for code management

- **Trello** for task tracking
- **Notion** for planning documents

Development Timeline

Milestone	**Task**	**Deadline**	**Responsible**
Milestone 1	Concept & Planning	14 April	Both
Milestone 2	API Integration & Basic UI	20 April	Both
Milestone 3	Database Setup & Linking	26 April	Member A
Milestone 4	UI Improvements & Login System	30 April	Member B
Milestone 5	Recommendation Engine	3 May	Member A
Milestone 6	Testing and Final Deployment	7 May	Both